

Minimax Criterion of Similarity for Video Information Processing

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Abstract — In this paper we propose new family of functions similarity allowing with a high degree of accuracy to process a video information. These functions forms an integral similarity estimate based on sequential minimax analysis image elements. Results of experimental researches for object detection in image and image sequences are presented.

Index Terms — Minimax similarity functions, object detection.

I. INTRODUCTION

For frame comparison of a video information a row of measures are used [1, 2] as unit for measurement of similarity images. Functions of similarity are applied for decision of some practical problems in a video information processing: positioning of images, objects detection on static images, pattern recognition, moving object detection [1, 3]. Normalized correlation function is widely used among known measures of similarity.

However, the problem of perfection the estimation methods of objects similarity is rather actual, because correlation characteristics of video images are far from ideal, i.e., and characterized by a significant level of secondary spikes and main spike inaccuracy [2,4]. It leads to false identifications of object, or ambiguity of positioning object on the image. Similar problems are existed for both at processing low-contrast gray-scale images, and in case of noise components presence on the image.

In process of algorithms perfection and expansion fields of images processing the correlation coefficient has undergone essential modifications, which have allowed generating on its basis a row of methods measures of similarity differing on properties and character-

istics.

In work [2] attempt to detailed analysis of existing methods for measuring various signal parameters to generate steady against various influences algorithms of objects similarity evaluation is undertaken. For this purpose all existing methods of similarity estimation classified on five groups: cross-correlation measures; classical statistics measures; derivative measures; ordinal measures; robust measures.

The analysis of the considered methods testifies that it is possible to speak only about quasi optimum of the considered similarity evaluation algorithms, depending on external conditions and type of analyzed data. Practically for all methods the basic problem is an accuracy of positioning, which is limited by the base width of the main correlation peak and the presence of intensive level of secondary spikes for the analysis of the image in a mix with noise [2, 5]. Except for this, it is necessary to note computing complexity problems, which arise in some cases even for images of small size, limiting dynamics of the analyzed image.

In this work, we present a new effective similarity criterion and family of function similarity for image and video processing. These functions forms an integral similarity estimate based on sequential minimax analysis image elements.

II. NEW SIMILARITY CRITERION

In this section, we present five minimax functions similarity for two images: images A , $N_1 \times N_2$ size, with elements a_{ij} and images B , $N_1 \times N_2$ size, with elements b_{ij} .

Multiplicative minimax similarity function R^M :

$$R^M = \prod_{i=0}^{N_1-1} \prod_{j=0}^{N_2-1} \frac{\min(a_{ij}, b_{ij})}{\max(a_{ij}, b_{ij})}.$$

Additive minimax similarity function R^S :

$$R^S = \frac{1}{N_1 N_2} \sum_{i=0}^{N_1-1} \sum_{j=0}^{N_2-1} \frac{\min(a_{ij}, b_{ij})}{\max(a_{ij}, b_{ij})}.$$

P-degree additive minimax similarity function R^{SD^p} :

$$R^{SD^p} = \frac{1}{N_I N_2} \sum_{i=0}^{N_I-1} \sum_{j=0}^{N_2-1} \frac{\min(a_{ij}^p, b_{ij}^p)}{\max(a_{ij}^p, b_{ij}^p)}.$$

For reducing the offered function sensitivity to a level of a direct bias of a signal it is necessary to use the average additive modification $\overline{R^s}$:

$$\left\{ \begin{array}{l} \text{if } |a_{ij} - \overline{a_{ij}}| > |b_{ij} - \overline{b_{ij}}|: \quad \overline{R^s} = \frac{1}{N_I N_2} \sum_{i=0}^{N_I-1} \sum_{j=0}^{N_2-1} \frac{b_{ij} - \overline{b_{ij}}}{a_{ij} - \overline{a_{ij}}}; \\ \text{else :} \quad \quad \quad \overline{R^s} = \frac{1}{N_I N_2} \sum_{i=0}^{N_I-1} \sum_{j=0}^{N_2-1} \frac{a_{ij} - \overline{a_{ij}}}{b_{ij} - \overline{b_{ij}}}. \end{array} \right.$$

where

$$\overline{a_{ij}} = \frac{1}{N_I N_2} \sum_{i=0}^{N_I-1} \sum_{j=0}^{N_2-1} a_{ij}, \quad \overline{b_{ij}} = \frac{1}{N_I N_2} \sum_{i=0}^{N_I-1} \sum_{j=0}^{N_2-1} b_{ij}.$$

Simple minimax similarity function R^{SM} :

$$R^{SM} = \min_{i,j \in 0, N-1} \left(\frac{\min(a_{ij}, b_{ij})}{\max(a_{ij}, b_{ij})} \right)$$

For the proposed function following basic properties for images A, B are obvious:

$$R^M(A, B) = 1, \Leftrightarrow A = B, \quad R^S(A, B) = 1, \Leftrightarrow A = B;$$

$$R^M(A, B) = R^M(B, A), \quad R^S(A, B) = R^S(B, A);$$

$$R^M(A, B) = 0, \Leftrightarrow a_{ij} = 0 \text{ } \forall b_{ij} = G,$$

$$R^S(A, B) = 0, \Leftrightarrow a_{ij} = 0 \text{ } \forall b_{ij} = G.$$

where G is a range of admissible brightness values.

III. COMPUTATIONAL COMPLEXITY

For calculation of the minimax similarity function of two frames for the image with size $N \times N$ pixels, it is required to perform N^2 operations of comparison, N^2 operations of division and $(N^2 - 1)$ opera-

tions of addition, and number of calculation is about $3N^2$ flops. For calculation of correlation function of two frames for the image size $N \times N$ pixels, we needs to spend $3N^2 + 1$ operations of multiplication, $3(N^2 - 1)$ operations of addition and one operation of division, i.e., number of calculation is evaluated by $6N^2$ size. Approximately $7N^2$ elementary calculation needs for function on the basis of the weighed sum of squared difference. Thus, in comparison with function of normalized correlation, the offered minimax function also provides reduction of calculation complexity, as min twice.

As result, minimax similarity functions can be used of search of objects on the static image and for object detecting moving objects on video sequences.

IV. EXPERIMENTAL RESULTS

4.1. Object detection

Search of objects on the image by a method of comparison to the sample is one of the basic detection methods [1]. Here the sample is compared to all objects, which are being on the image, by in series moving along the image, as a rule, from left to right, from top to down. As a guideline of detection one of known similarity function is used.

For an estimation of qualitative characteristics of offered minimax similarity function and its modifications, experiments on searching gray-scale objects on images are lead. In figures 2-4 functions of correlation type and minimax similarity functions for object and images, which are presented in fig. 1, are shown.

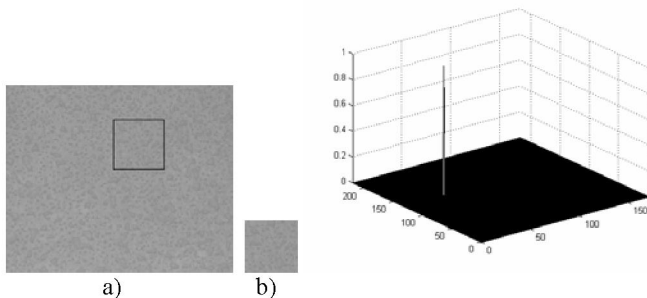


Fig. 1: a) Image; b) Object

Fig. 2 Similarity function R^M

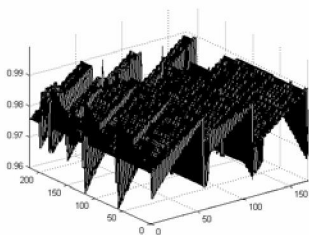


Fig. 3 Normalized cross-correlation

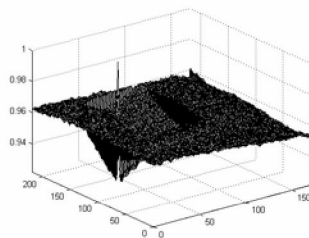
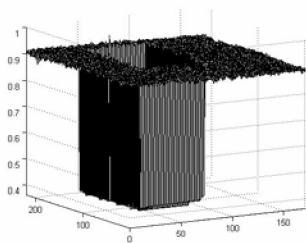
Fig. 4 Similarity function R^S 

Fig. 5 Similarity function on the basis of Hausdorff metrics

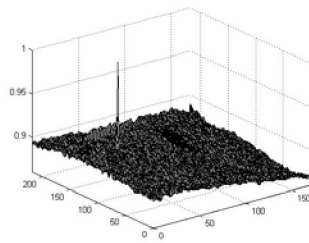
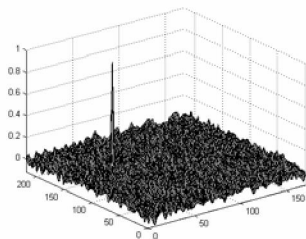
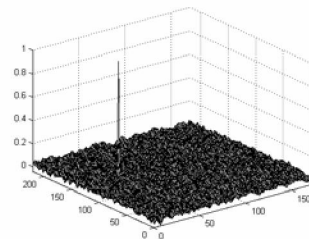
Fig.6 Similarity function R^{SD^3} 

Fig. 7 Normalized averaged cross-correlation

Fig.8 Similarity function $\overline{R^S}$

The analysis of result indicates the raised resolution of the offered minimax similarity function. The most qualitative, from this point of view, is multiplicative minimax similarity function. However, it is necessary to note, that the given modification of the offered

function, providing exact localization and high contrast of function at images processing, is characterized by a significant sensitivity to noise influence. At presence of noise it is possible to use the proposed modifications of minimax similarity function: additive and averaged additive.

4.2. Detection of moving objects

One of the basic methods of mobile objects detection on dynamic images is the method, that uses level of similarity for comparison of corresponding fragments of the video sequence adjoining frames [3]. If the similarity function does not exceed the preset threshold, the decision on presence of changes for analyzable fragment of the frame, is made. For do it, as a measure of similarity, the correlation coefficient is widely used. Figure 9 shows results of mobile object detection.

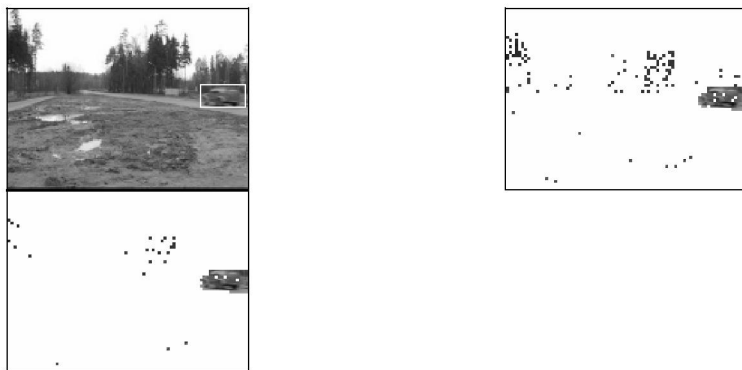


Fig. 9 Detection of moving object result: a) mobile object; b) extracted object on the basis of correlation; c) extracted object on the basis of minimax additive function

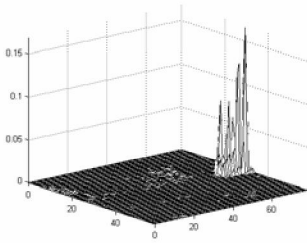
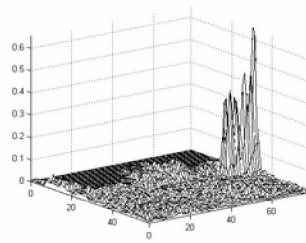


Fig. 10 Normalized cross-correlation

Fig.11 Similarity function R^s

Presented in fig.10-11 graphs of a normalized cross-correlation and additive minimax similarity function indicate higher level of proposed function contrast for localization of mobile objects on dynamic images.

V. CONCLUSION

New minimax criterions family for image and video processing is developed. Set of functions includes: multiplicative and additive minimax similarity function, simple minimax similarity function, p-degree additive minimax function and average additive modification.

Results of researches testify that the proposed family of minimax similarity function can provide the improved resolution for processing static and dynamic images, in comparison with known measures of similarity. For sensitivity increase to noise it is necessary to use additive minimax similarity function. As a result, the offered family of minimax similarity function, possessing the advanced qualitative characteristics, allows increasing efficiency of static and dynamic images processing methods as a whole.

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